VIT-AP UNIVERSITY, ANDHRA PRADESH

Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech **Date:** 26/2/2026
Faculty Name: Prof. S.Gopikrishnan **School:** SCOPE
Student name: shanmukh Malladi **Reg. no.:** 23BCE7317

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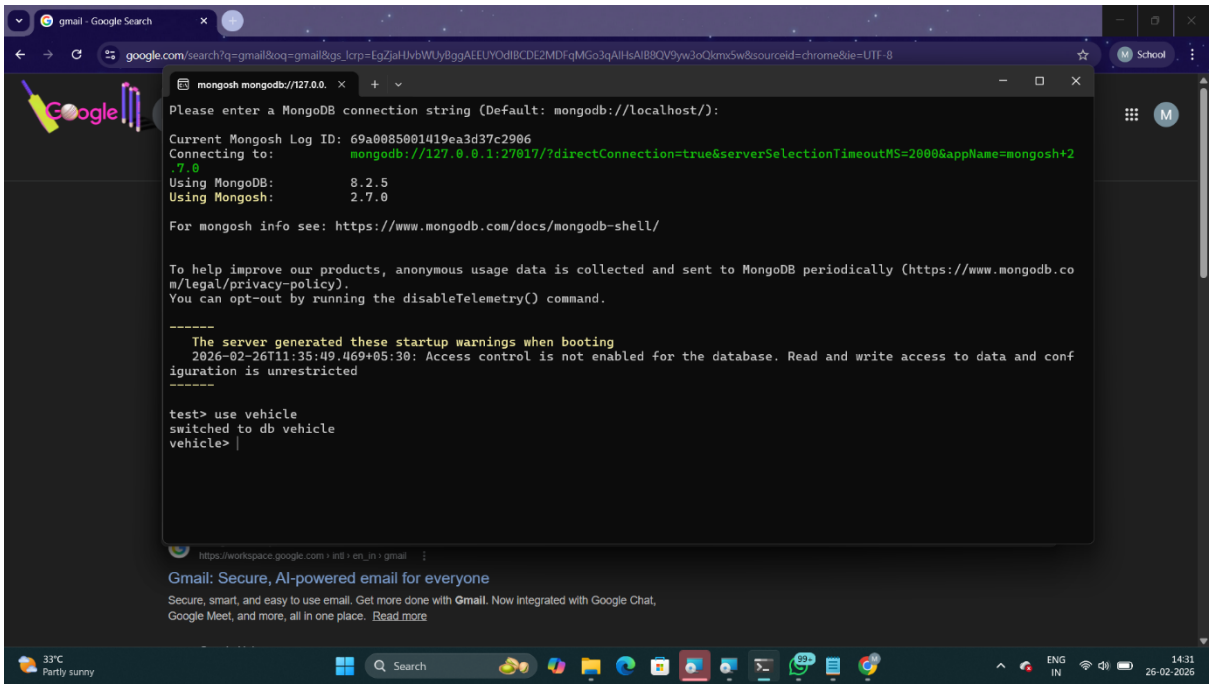
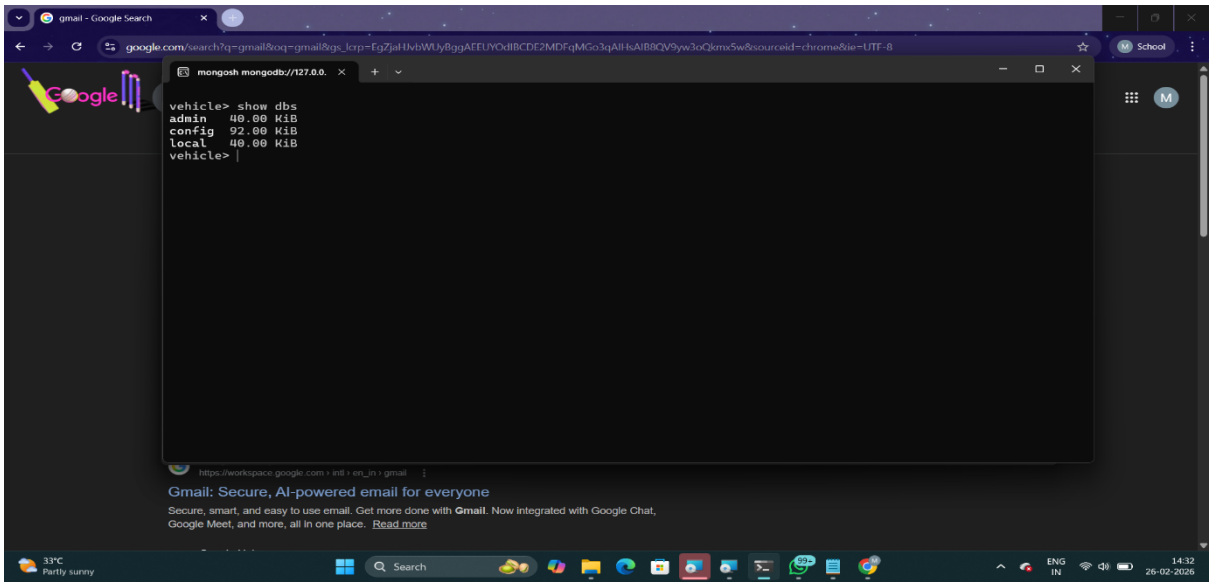
1. Use MongoDB to implement the following DB operations

   1. Create a database called 'vehicles' and *write* a MongoDB query to select database as "vehicles".



   2. Write a MongoDB query to display all the databases.



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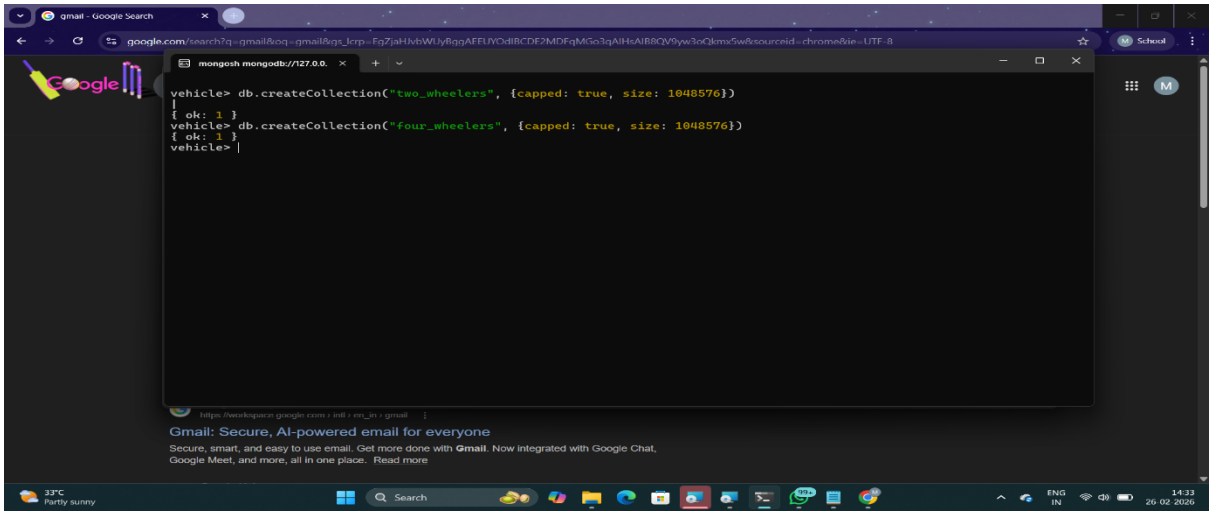
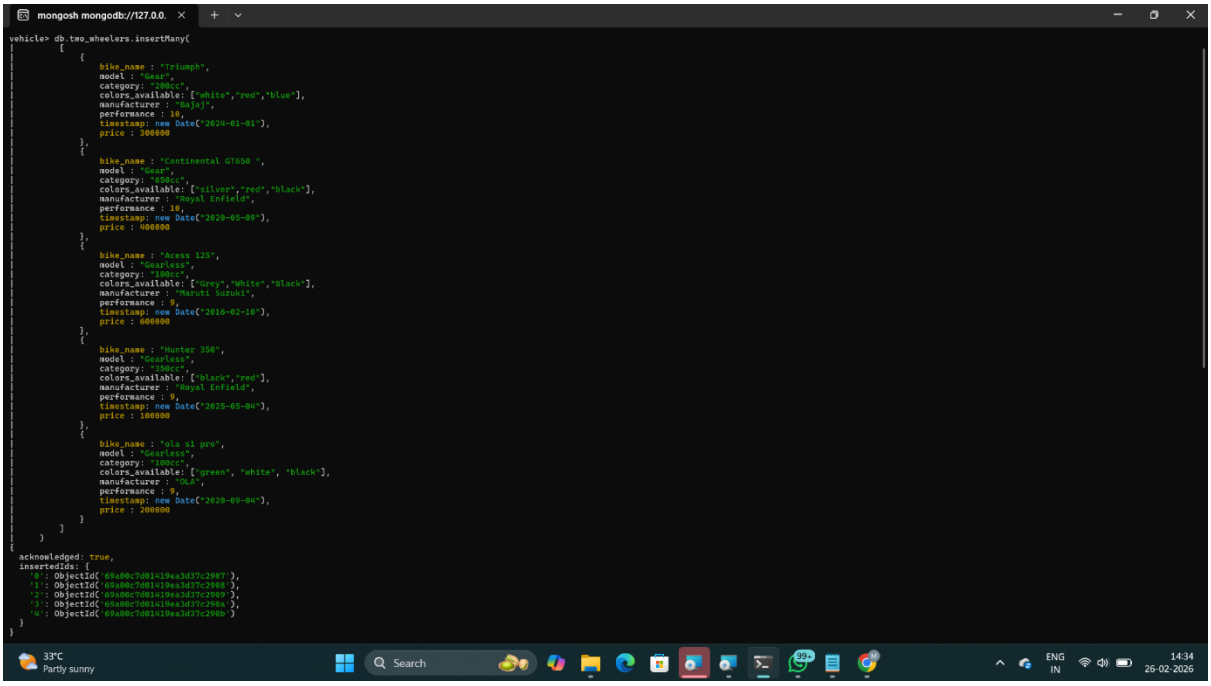
3. Create a collection called 'two_wheelers'. (use capping) and Create a collection called 'four_wheelers'.



4. Add 5 two-wheeler details to the collection named 'two_wheelers'. Each document consists of following fields as bike_name, model (gear or gearless), category (100cc, 125cc, 150cc, 200cc), colors_available (red, black, blue, sport red etc) as array, manufacturer, performance (out of 10), timestamp (date and year release) and price.



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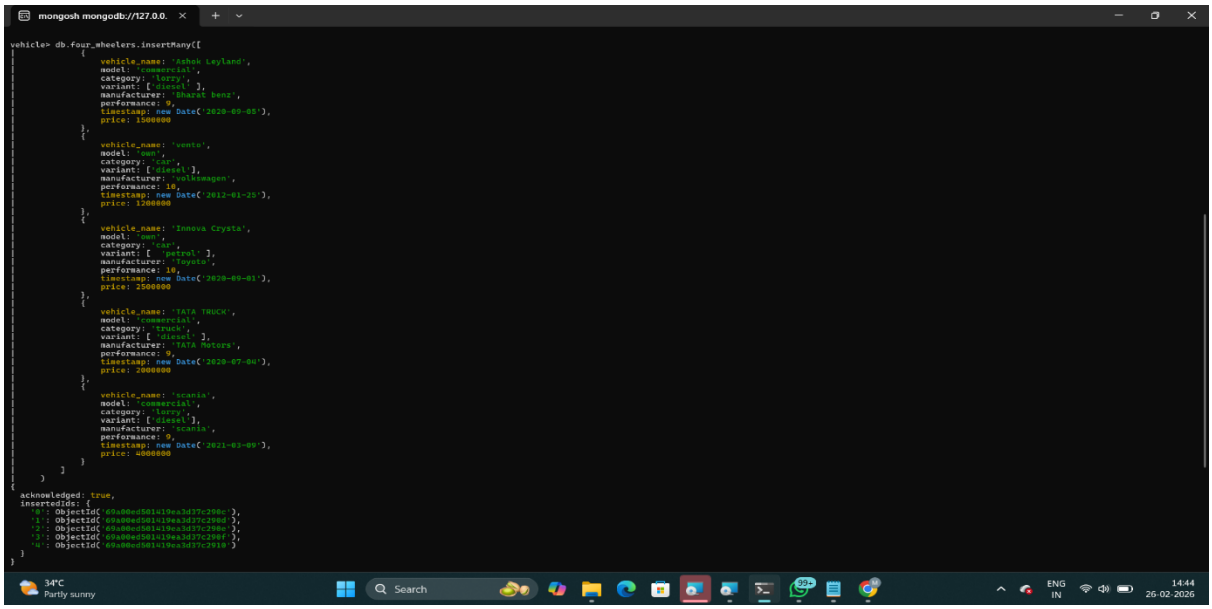
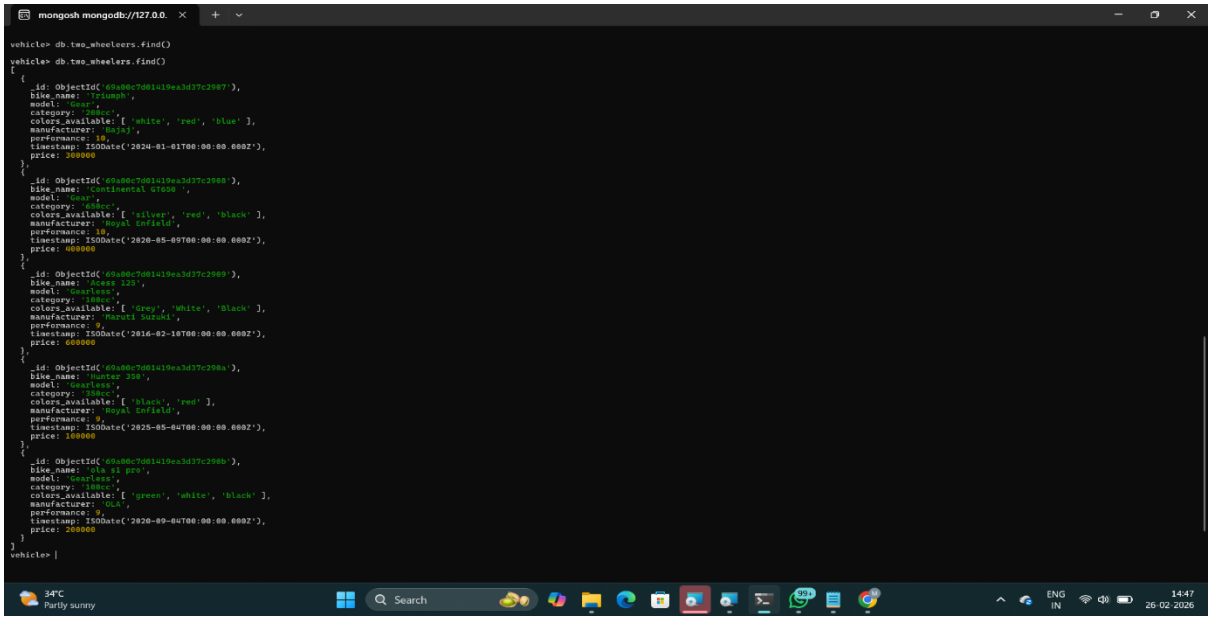
5. Add 5 four-wheeler details to the collection named 'four_wheelers'. Each document consists of following fields as vehicle_name, model (commercial or own), category (car, lorry, bus, mini truck, heavy truck, containers), variants (vxi, zxi, petrol, diesel etc) as array, manufacturer, performance (out of 10), timestamp (date and year release) and price.



6. Write a MongoDB query to display all documents available in two_wheelers and four_wheelers.



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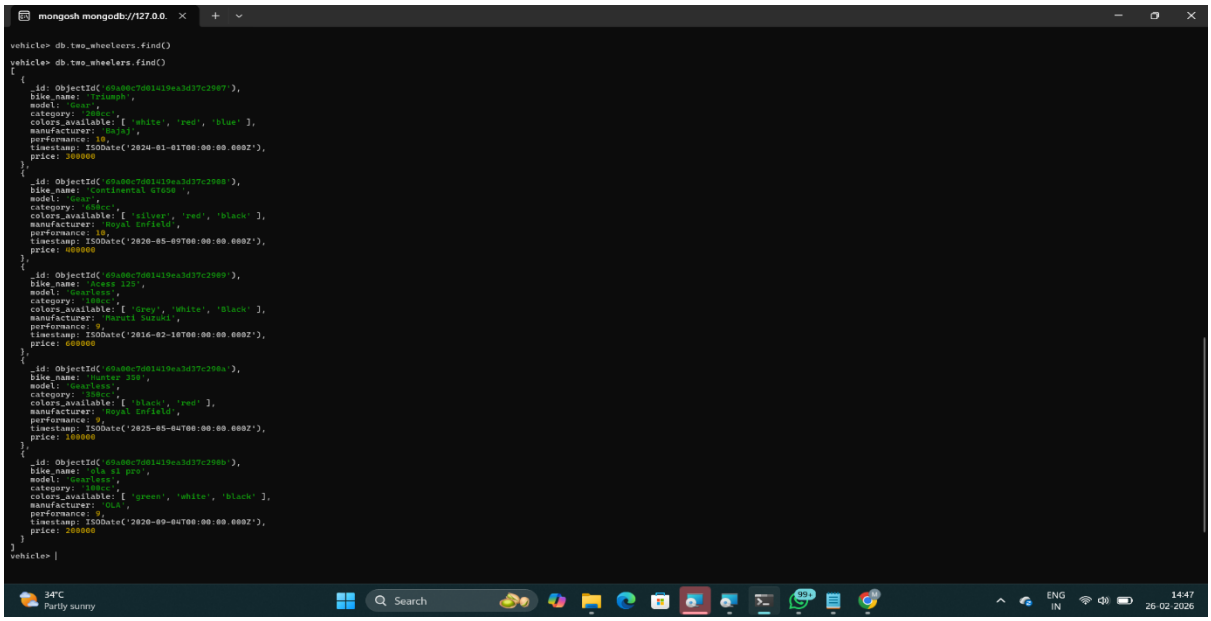
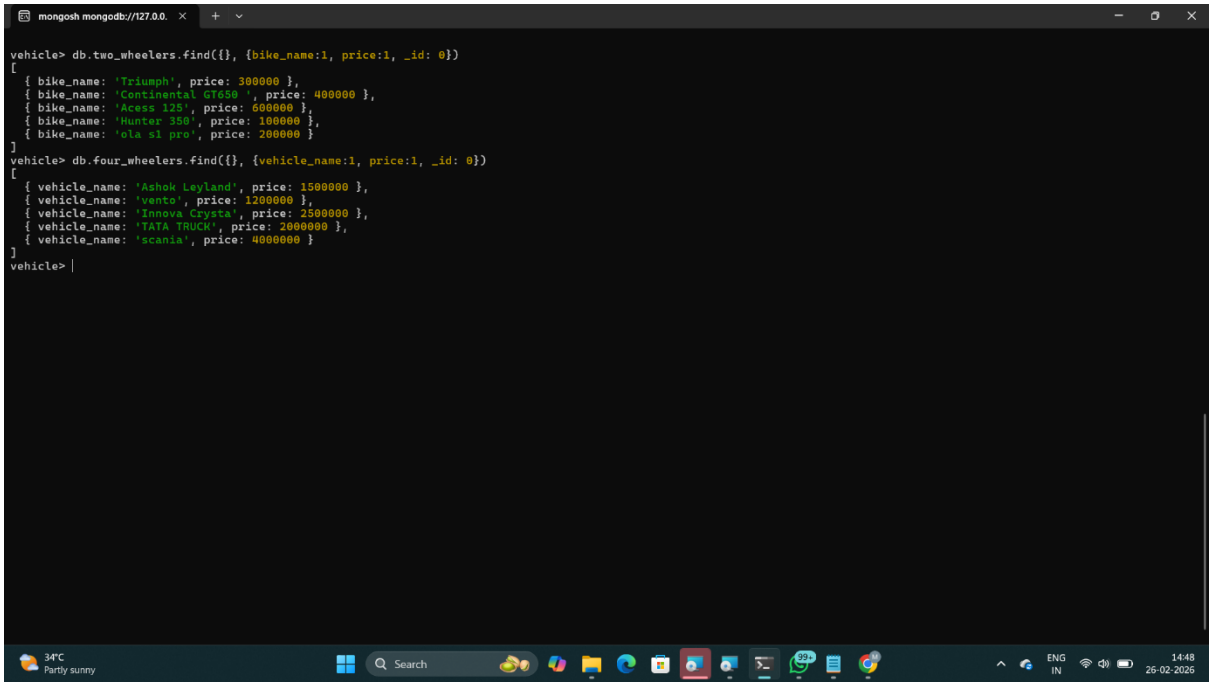


7. Write a MongoDB query to display only vehicle name and price in all the collection of the database



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8. Write a MongoDB query to display two_wheelers from a particular company

```
vehicle> db.two_wheelers.find({manufacturer: "Royal Enfield"}, {})
[
  {
    _id: ObjectId('69a00c7d01419ea3d37c2908'),
    bike_name: 'Continental GT650 ',
    model: 'Gear',
    category: '650cc',
    colors_available: [ 'silver', 'red', 'black' ],
    manufacturer: 'Royal Enfield',
    performance: 10,
    timestamp: ISODate('2020-05-09T00:00:00.000Z'),
    price: 400000
  },
  {
    _id: ObjectId('69a00c7d01419ea3d37c290a'),
    bike_name: 'Hunter 350',
    model: 'Gearless',
    category: '350cc',
    colors_available: [ 'black', 'red' ],
    manufacturer: 'Royal Enfield',
    performance: 9,
    timestamp: ISODate('2025-05-04T00:00:00.000Z'),
    price: 100000
  }
]
vehicle>
```

9. Write a MongoDB query to display four_wheelers available in diesel variants

```
vehicle> db.four_wheelers.find({variant: {$in: ["diesel"]}}, {})
[
  {
    _id: ObjectId('69a00ed501419ea3d37c290c'),
    vehicle_name: 'Ashok Leyland',
    model: 'commercial',
    category: 'lorry',
    variant: [ 'diesel' ],
    manufacturer: 'Bharat benz',
    performance: 9,
    timestamp: ISODate('2020-09-05T00:00:00.000Z'),
    price: 1500000
  },
  {
    _id: ObjectId('69a00ed501419ea3d37c290d'),
    vehicle_name: 'vento',
    model: 'own',
    category: 'car',
    variant: [ 'diesel' ],
    manufacturer: 'volkswagen',
    performance: 10,
    timestamp: ISODate('2012-01-25T00:00:00.000Z'),
    price: 1200000
  },
  {
    _id: ObjectId('69a00ed501419ea3d37c290f'),
    vehicle_name: 'TATA TRUCK',
    model: 'commercial',
    category: 'truck',
    variant: [ 'diesel' ],
    manufacturer: 'TATA Motors',
    performance: 9,
    timestamp: ISODate('2020-07-04T00:00:00.000Z'),
    price: 2000000
  },
  {
    _id: ObjectId('69a00ed501419ea3d37c2910'),
    vehicle_name: 'scania',
    model: 'commercial',
    category: 'lorry',
    variant: [ 'diesel' ],
    manufacturer: 'scania',
    performance: 9,
    timestamp: ISODate('2021-03-09T00:00:00.000Z'),
    price: 4000000
  }
]
vehicle>
```

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Student name: shanmukh Malladi **Reg. no.:** 23BCE7317

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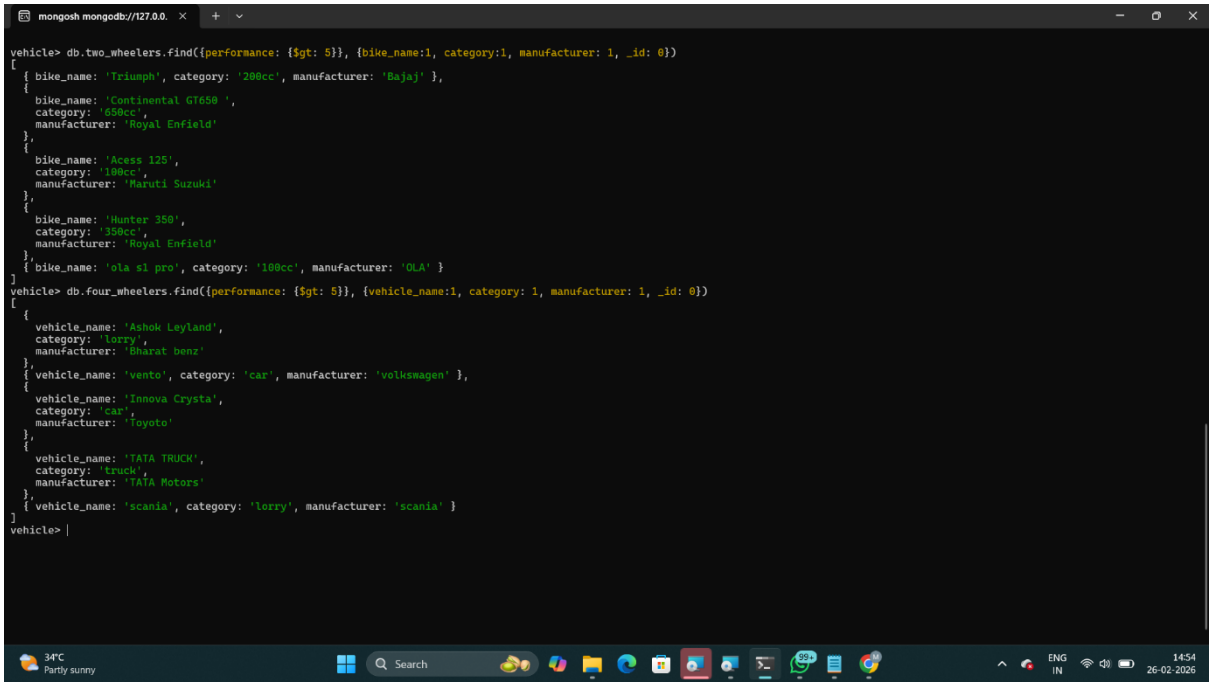
10. Write a MongoDB query to display vehicles name, category and manufacturer details whose rating is more than 5.



```
vehicle> db.two_wheelers.find({performance: {$gt: 5}}, {bike_name:1, category:1, manufacturer: 1, _id: 0})
[
  { bike_name: 'Triumph', category: '200cc', manufacturer: 'Bajaj' },
  {
    bike_name: 'Continental GT650 ',
    category: '650cc',
    manufacturer: 'Royal Enfield'
  },
  {
    bike_name: 'Acess 125',
    category: '100cc',
    manufacturer: 'Maruti Suzuki'
  },
  {
    bike_name: 'Hunter 350',
    category: '350cc',
    manufacturer: 'Royal Enfield'
  },
  { bike_name: 'ola s1 pro', category: '100cc', manufacturer: 'OLA' }
]
vehicle> db.four_wheelers.find({performance: {$gt: 5}}, {vehicle_name:1, category: 1, manufacturer: 1, _id: 0})
[
  {
    vehicle_name: 'Ashok Leyland',
    category: 'lorry',
    manufacturer: 'Bharat benz'
  },
  { vehicle_name: 'vento', category: 'car', manufacturer: 'volksmagen' },
  {
    vehicle_name: 'Innova Crysta',
    category: 'car',
    manufacturer: 'Toyoto'
  },
  {
    vehicle_name: 'TATA TRUCK',
    category: 'truck',
    manufacturer: 'TATA Motors'
  },
  { vehicle_name: 'scania', category: 'lorry', manufacturer: 'scania' }
]
vehicle>
```

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Student name: shanmukh Malladi **Reg. no.:** 23BCE7317

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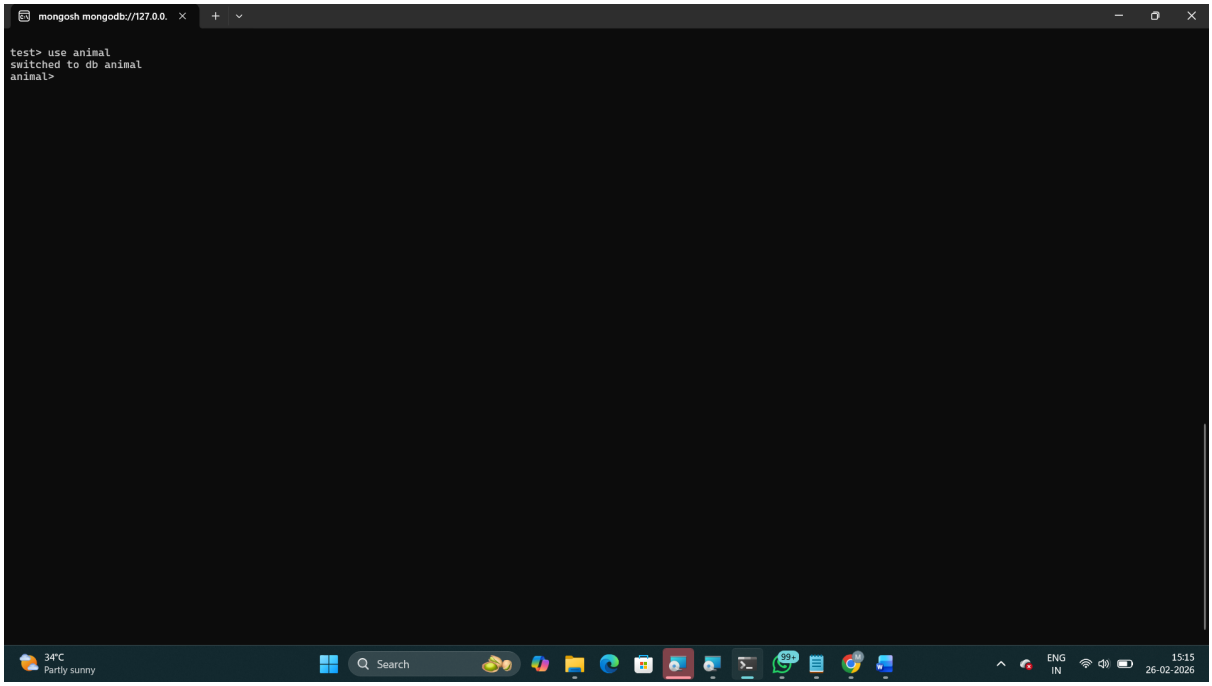
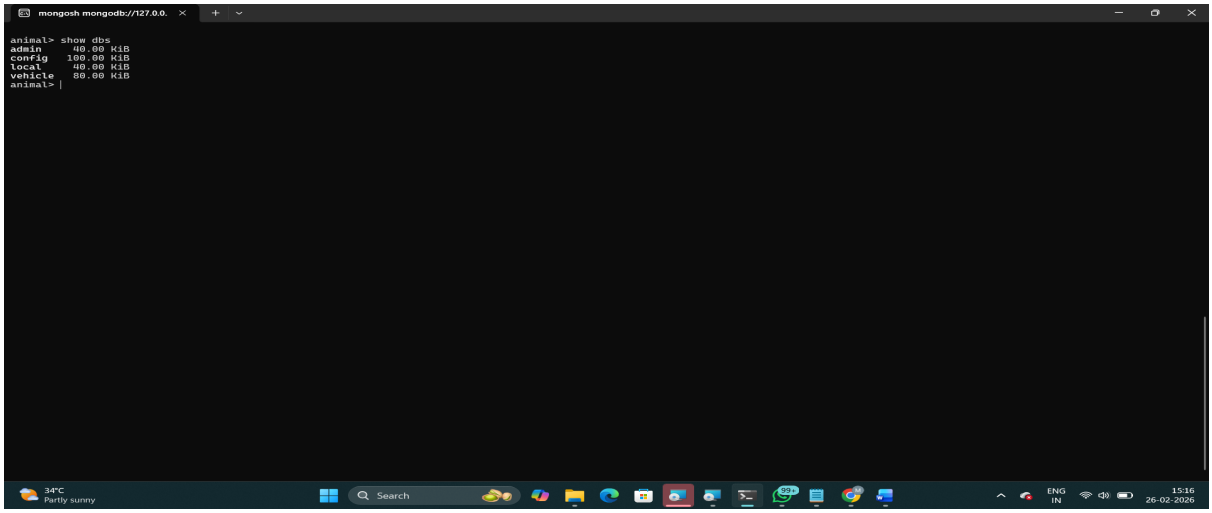
2. Use MongoDB to implement the following DB operations for a Zoo

1. Create a database called 'animal' and *write* a MongoDB query to select database as 'animal'.



2. Write a MongoDB query to display all the databases.



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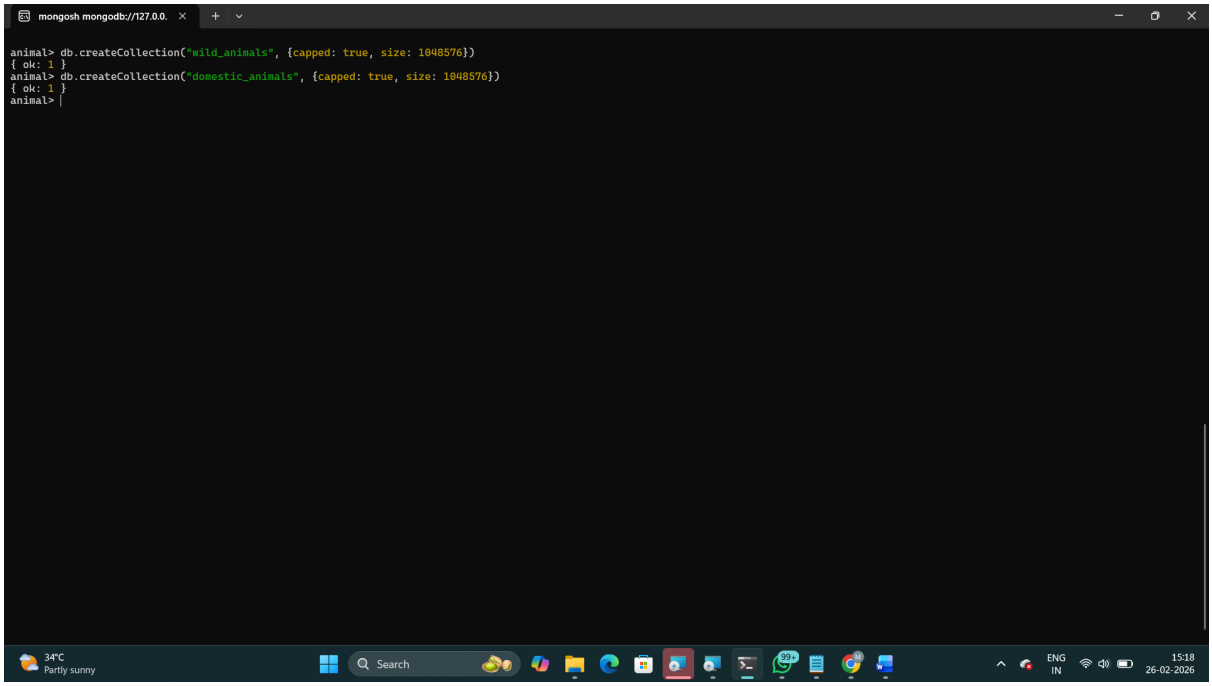
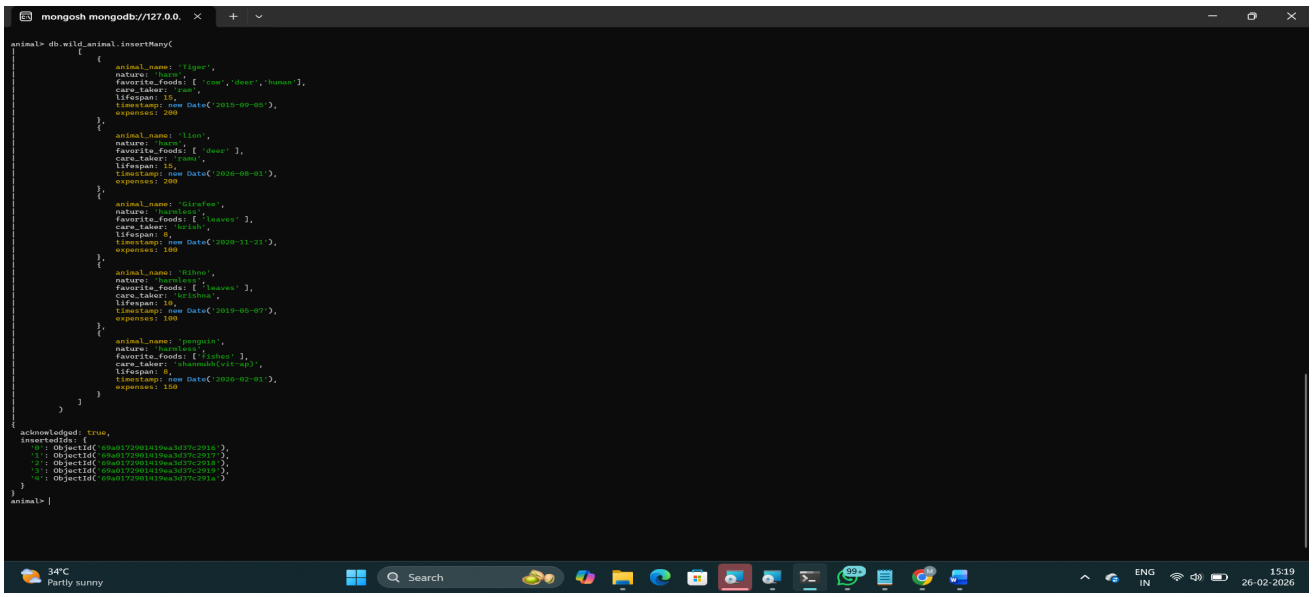
3. Create a collection called 'wild_animals'.(use capping) and Create a collection called 'domestic_animals'.



4. Add 5 wild_animal details to the collection named 'wild_animals'. Each document consists of following fields as animal_name, nature (harm or harmless), favorite_foods (meat, rabbits, deer etc) as array, care_taker_name, life span (in years), timestamp (when the animal registered at the Zoo) and expenses.



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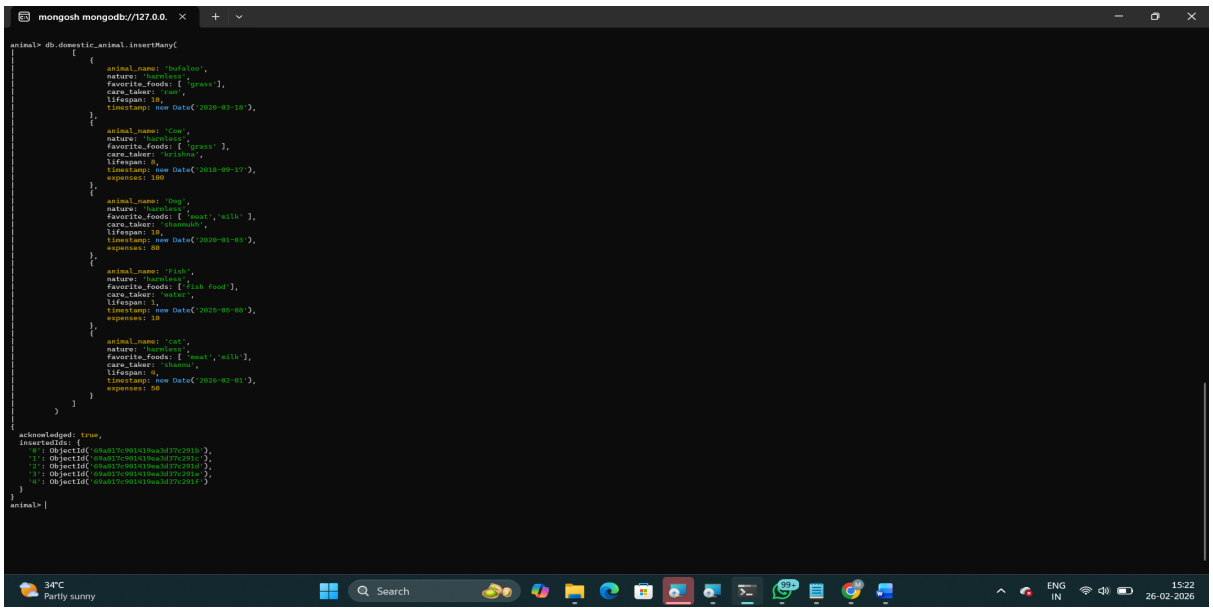
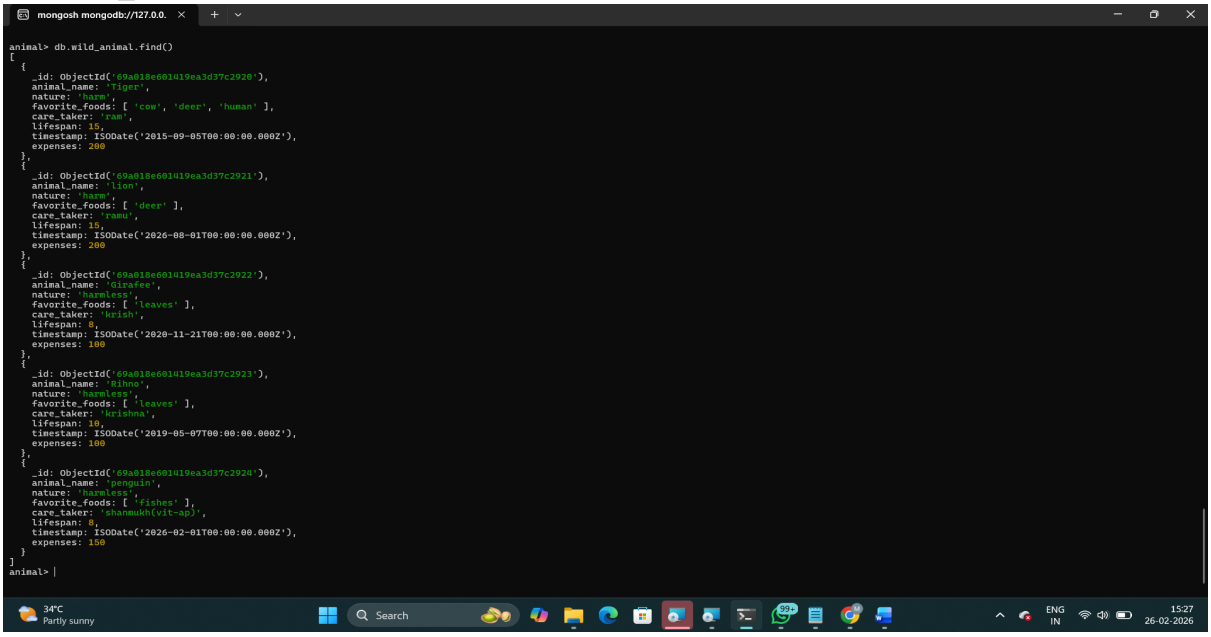
5. Add 5 domestic-animal details to the collection named 'domestic_animals'. Each document consists of following fields as animal_name, gender (male or female), favorite_foods (meat, rabbits, deer etc) as array, animal_petname, life span (in years), timestamp (when the animal registered at the Zoo) and expenses.



6. Write a MongoDB query to display all documents available in wild_animals and domastic_animals.



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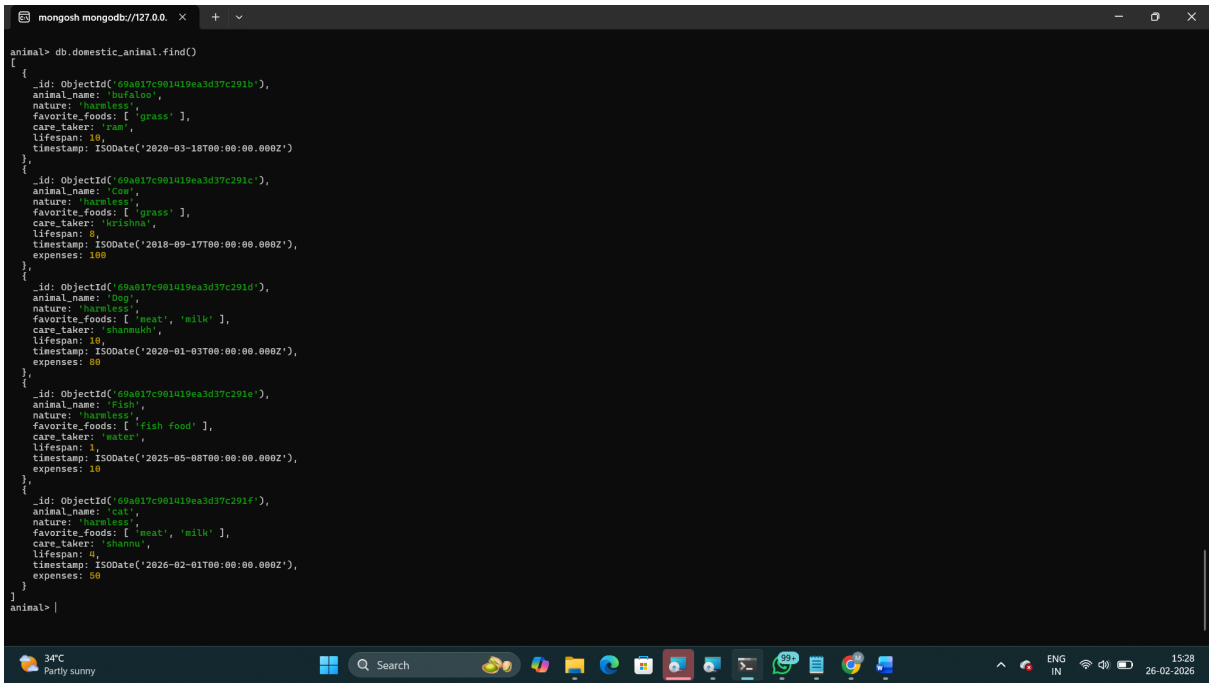
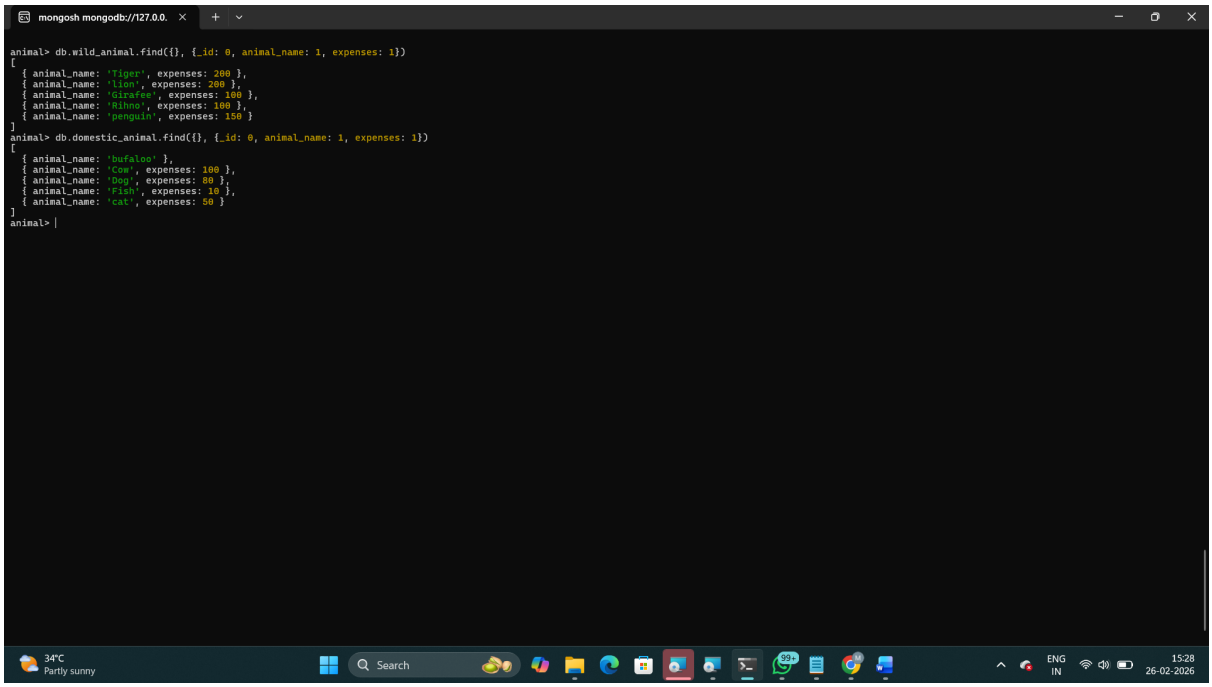


```
mongosh mongodb://127.0.0.

animal> db.domestic_animal.find()
[
  {
    _id: ObjectId('69a017c901419ea3d37c291b'),
    animal_name: 'bufaloo',
    nature: 'harmless',
    favorite_foods: [ 'grass' ],
    care_taker: 'ram',
    lifespan: 10,
    timestamp: ISODate('2020-03-18T00:00:00.000Z')
  },
  {
    _id: ObjectId('69a017c901419ea3d37c291c'),
    animal_name: 'Cow',
    nature: 'harmless',
    favorite_foods: [ 'grass' ],
    care_taker: 'krishna',
    lifespan: 8,
    timestamp: ISODate('2018-09-17T00:00:00.000Z'),
    expenses: 100
  },
  {
    _id: ObjectId('69a017c901419ea3d37c291d'),
    animal_name: 'Dog',
    nature: 'harmless',
    favorite_foods: [ 'meat', 'milk' ],
    care_taker: 'shanmukh',
    lifespan: 10,
    timestamp: ISODate('2020-01-03T00:00:00.000Z'),
    expenses: 80
  },
  {
    _id: ObjectId('69a017c901419ea3d37c291e'),
    animal_name: 'Fish',
    nature: 'harmless',
    favorite_foods: [ 'fish food' ],
    care_taker: 'water',
    lifespan: 1,
    timestamp: ISODate('2025-05-08T00:00:00.000Z'),
    expenses: 10
  },
  {
    _id: ObjectId('69a017c901419ea3d37c291f'),
    animal_name: 'cat',
    nature: 'harmless',
    favorite_foods: [ 'meat', 'milk' ],
    care_taker: 'shannu',
    lifespan: 4,
    timestamp: ISODate('2026-02-01T00:00:00.000Z'),
    expenses: 50
  }
]
animal>
```

7. Write a MongoDB query to display only animal name and expenses in all the collection of the database



```
mongosh mongodb://127.0.0.

animal> db.wild_animal.find({}, {_id: 0, animal_name: 1, expenses: 1})
[
  { animal_name: 'Tiger', expenses: 200 },
  { animal_name: 'Lion', expenses: 200 },
  { animal_name: 'Girafee' , expenses: 100 },
  { animal_name: 'Rihno', expenses: 100 },
  { animal_name: 'penguin', expenses: 150 }
]
animal> db.domestic_animal.find({}, {_id: 0, animal_name: 1, expenses: 1})
[
  { animal_name: 'bufaloo' },
  { animal_name: 'Cow', expenses: 100 },
  { animal_name: 'Dog', expenses: 80 },
  { animal_name: 'Fish', expenses: 10 },
  { animal_name: 'cat', expenses: 50 }
]
animal>
```

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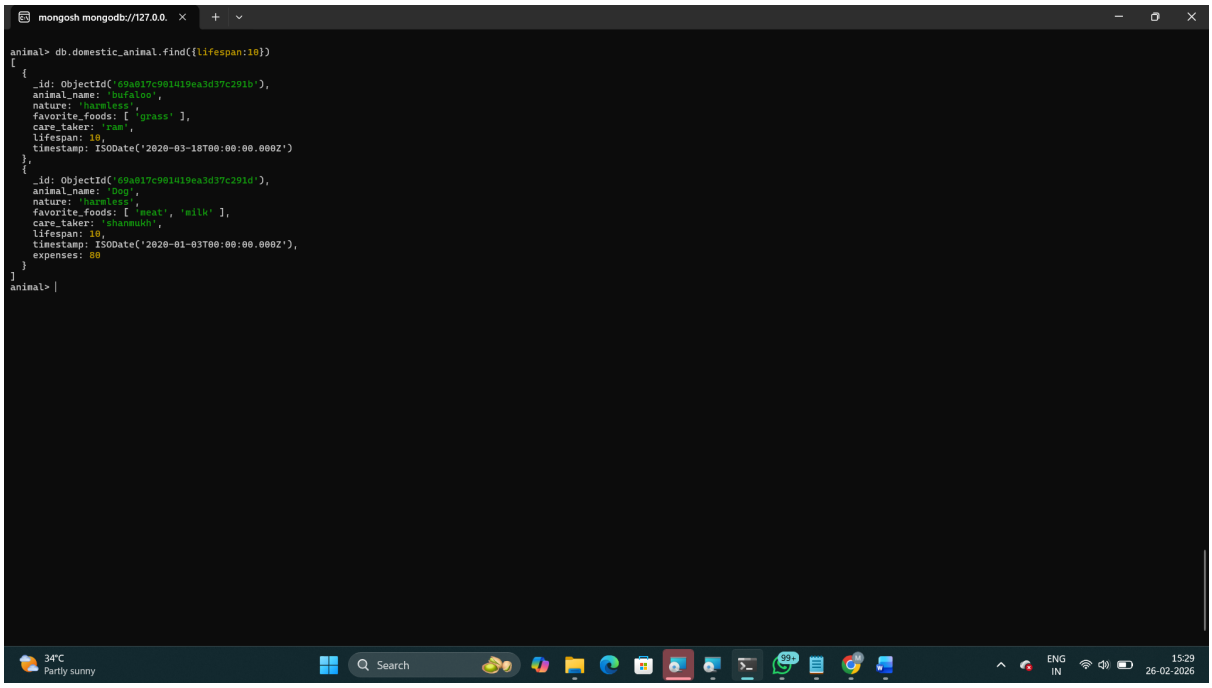
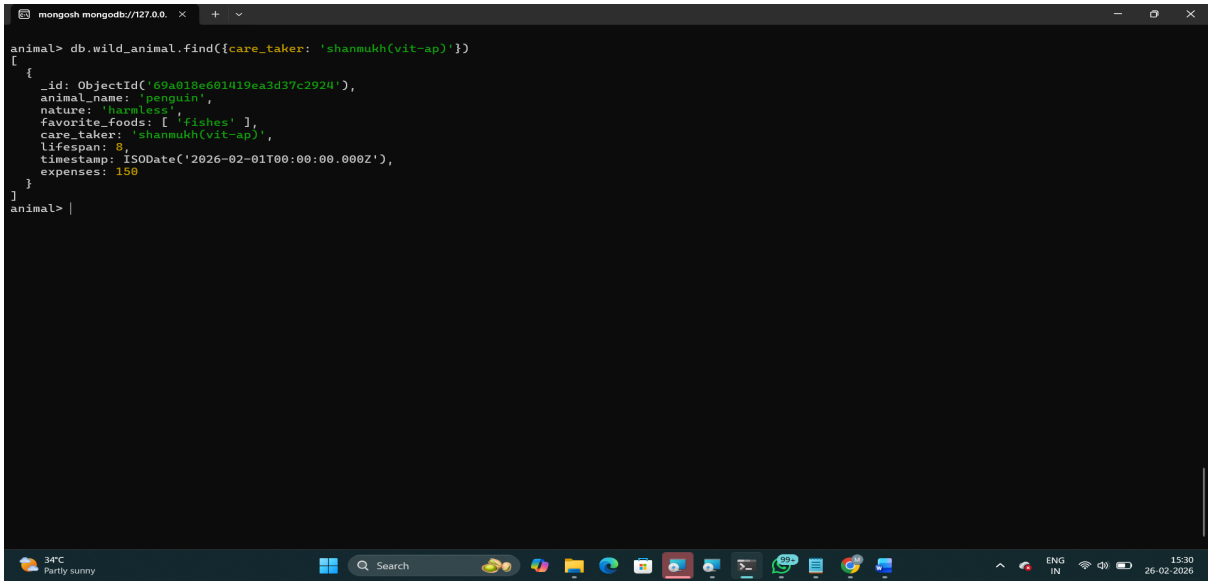
8. Write a MongoDB query to display domestic_animals whose life is a particular year\



9. Write a MongoDB query to display wild_animals available under a particular care_taker



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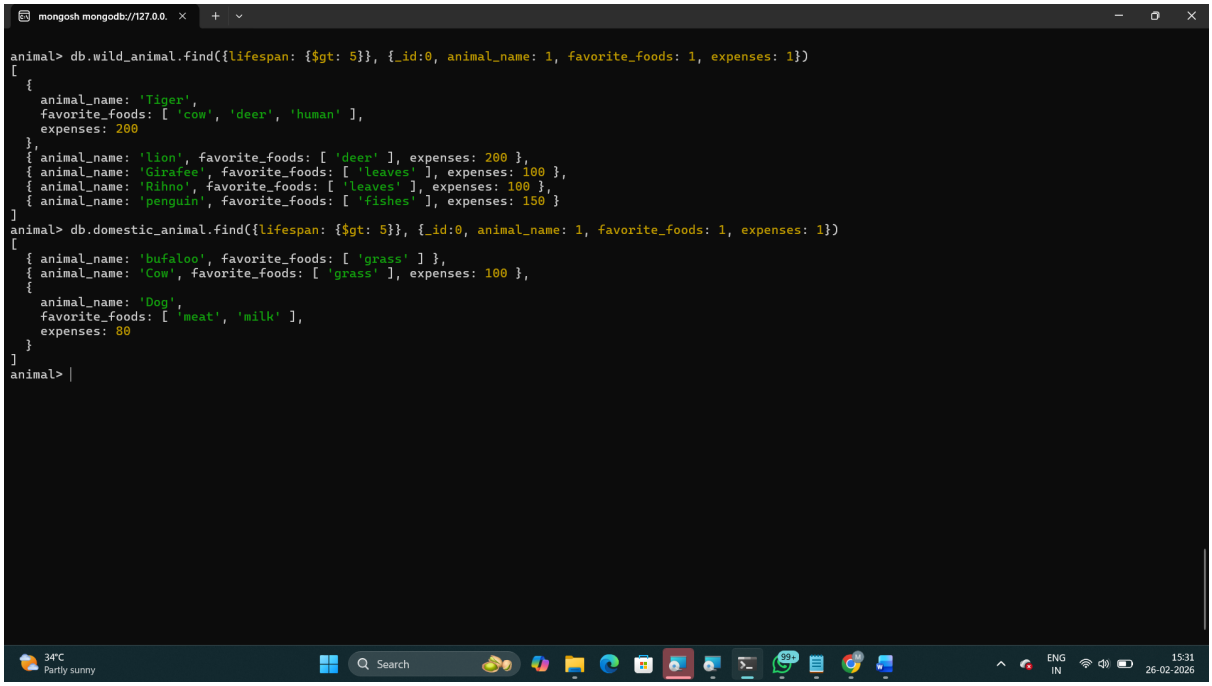
10. Write a MongoDB query to display animal name, favorite_foods and expenses details whose lifespan is more than 5 years.



```
animal> db.wild_animal.find({lifespan: {$gt: 5}}, {_id:0, animal_name: 1, favorite_foods: 1, expenses: 1})
[
  {
    animal_name: 'Tiger',
    favorite_foods: [ 'cow', 'deer', 'human' ],
    expenses: 200
  },
  { animal_name: 'lion', favorite_foods: [ 'deer' ], expenses: 200 },
  { animal_name: 'Girafee', favorite_foods: [ 'leaves' ], expenses: 100 },
  { animal_name: 'Rihno', favorite_foods: [ 'leaves' ], expenses: 100 },
  { animal_name: 'penguin', favorite_foods: [ 'fishes' ], expenses: 150 }
]
animal> db.domestic_animal.find({lifespan: {$gt: 5}}, {_id:0, animal_name: 1, favorite_foods: 1, expenses: 1})
[
  { animal_name: 'bufaloo', favorite_foods: [ 'grass' ] },
  { animal_name: 'Cow', favorite_foods: [ 'grass' ], expenses: 100 },
  {
    animal_name: 'Dog',
    favorite_foods: [ 'meat', 'milk' ],
    expenses: 80
  }
]
animal>
```